

ENTHUSIAST, LEADER & ACHIEVER COURSE

PHASE : ALL ENTHUSIAST, MLA,B,K,M,P,Q,R,MAZA,B,C,D,E,F,G,
H,L,M,N,O,P,Q,R,U,V,MAAZ,AW,AX,AY,MAPA,PB,SPARK PRO,LAKSHYA

TARGET : PRE-MEDICAL 2025

Test Type : MAJOR

Test Pattern : NEET (UG)

TEST DATE : 08-04-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	2	4	2	4	3	2	4	2	3	4	2	4	3	1	2	2	4	1	3	3	3	3	3	2	2	4	4	3	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	3	1	3	3	4	3	3	3	4	3	2	2	3	4	2	3	3	4	3	3	3	3	4	1	2	3	4	1	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	2	1	4	2	2	1	1	3	3	4	4	2	4	1	1	4	2	1	2	3	4	2	2	3	1	3	4	3	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	4	4	2	4	2	4	4	1	1	1	1	3	2	1	3	4	2	2	3	3	4	1	4	4	4	4	4	3	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	2	3	4	4	4	4	1	1	3	4	1	3	2	4	3	1	1	1	3	1	2	4	2	4	1	2	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	4	4	1	4	4	4	4	3	2	1	4	1	2	3	2	2	3	1	3	2	1	4	1	3	3	3	4	3	4

HINT - SHEET

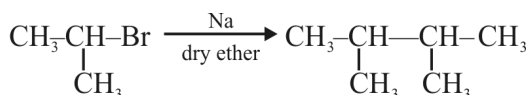
1. Ans (1)

Boiling point of Alkane

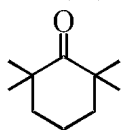
$\propto$ Mol. wt. (No. of C)

$\propto \frac{1}{\text{Branching}}$ (if mol. wt. same)

2. Ans (2)



3. Ans (4)



do not show tautomerism due to absence of α -H.

4. Ans (2)

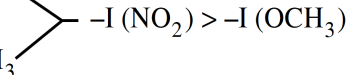
$$\text{Acidic strength} \propto \frac{-\text{M}, -\text{H}, -\text{I}}{+\text{M}, +\text{H}, +\text{I}}$$

(a) -M of $-\text{NO}_2$

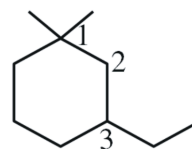
(b) -I of $-\text{NO}_2$

(c) No group

(d) -I of $-\text{OCH}_3$



6. Ans (3)



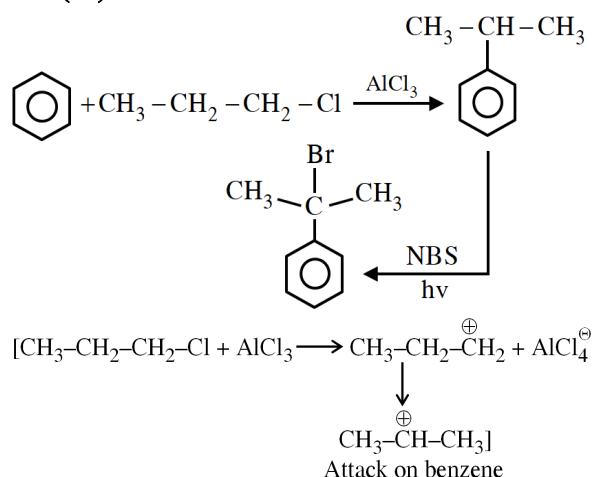
3-Ethyl-1,1-dimethylcyclohexane

7. Ans (2)

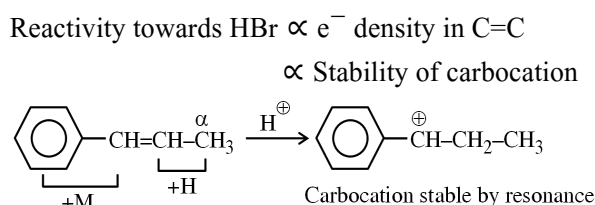
Functional group is different in both compound.

$\therefore$ functional group isomers.

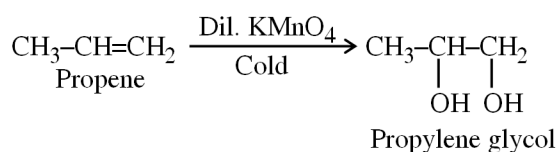
8. Ans (4)



9. Ans (2)

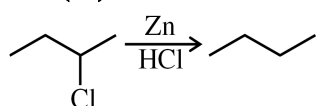


10. Ans (3)

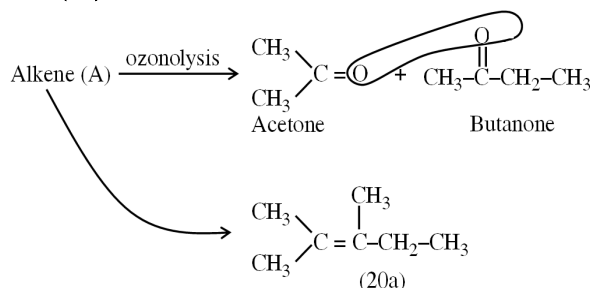


Propene and propyne cannot be distinguished by Baeyer's reagent because both give test of unsaturation.

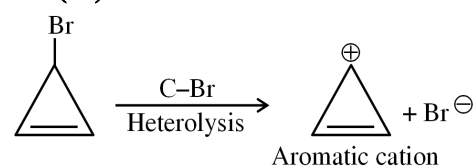
11. Ans (4)



12. Ans (2)

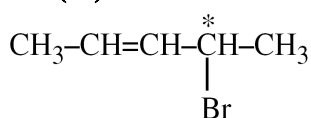


13. Ans (4)



15. Ans (1)

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It shows GI around double bond and show optical isomerism due to presence of chiral carbon.

16. Ans (2)

H2SO4 and HNO3 have Inter molecular H-bond.

17. Ans (2)

ICl4^- $\rightarrow$ square planer $\theta = 90^\circ$

XeF2 $\rightarrow$ Linear $\theta = 180^\circ$ (no effect of l.p. present in Bond angle.)

18. Ans (4)

NCERT-XI, Part-I Pg.131

20. Ans (3)

Be2, Ne2, PH5, BF6^{3-}

21. Ans (3)

LiF < Li2O < Li3N

covalent nature $\propto$ charge of anion

22. Ans (3)

Dipole-induced dipole interaction present in HCl & He atom. Because HCl molecule is polar and induces dipole in He atom.

23. Ans (3)

It is As $4s^2 4p^3$

25. Ans (2)

$\underbrace{P^{3-} > S^{2-}}_{3^{\text{rd}} \text{ period}} > \underbrace{N^{3-} > O^{2-} > Na^+ > Mg^{+2}}_{\text{Isoelectronic}}$

26. Ans (2)

I.E. F > Cl

EA Cl > F So I.E. Cl^- > F^-
 as I.E. (anion) = E.A. (neutral atom)

27. Ans (4)

EN order Cl $\approx$ N < O < F

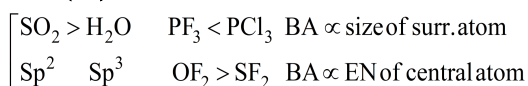
28. Ans (4)

acidic and amphoteric oxide can react with NaOH

29. Ans (3)

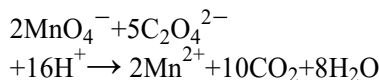
I3^+ bent, ICl4^- square planer
ClF3 T-shape, XeF2 linear

30. Ans (3)



31. Ans (1)

The balanced redox reaction is



33. Ans (1)

Understanding the Concepts

Le Chatelier's Principle: When a system at equilibrium is subjected to a change in concentration, temperature, or pressure, the system will shift in a direction to minimize

- the effect of the change.
- Endothermic Reaction ($\Delta H = +ve$): Heat is absorbed during the reaction. Increasing temperature favors the forward reaction.
- Effect of Pressure: Pressure changes primarily affect gaseous reactions. If the number of moles of gaseous products is less than the number of moles of gaseous reactants, increasing pressure favors the forward reaction.

Analyzing the Reaction

- $\text{X}_2(\text{g}) + \text{Y}_2(\text{g}) \rightleftharpoons \text{Z}_2(\text{g}); \Delta H = +ve$
The reaction is endothermic ($\Delta H = +ve$), so increasing temperature will favor the formation of Z_2 (the product).
The number of moles of gaseous reactants ($1 + 1 = 2$) is greater than the number of moles of gaseous products (1). Therefore, increasing pressure will favor the formation of Z_2 .

Determining the Favorable Conditions

To maximize the formation of the product

- (Z_2), we need:
 - High temperature (to favor the endothermic reaction).
 - High pressure (to favor the side with fewer moles of gas).

Answer

The correct answer is:

- 1. High temperature & High pressure

37. Ans (3)

$$\Delta H = \Delta E + \Delta n_g RT$$

$$\Delta H = -1364.47 + (-1)(8.314 \times 10^{-3}) (298)$$

$$\Delta H = -1366.95 \text{ kJ/mol}$$

38. Ans (3)

$$\lambda = \frac{h}{mv}$$

$$m = \frac{h}{\lambda \times v}$$

$$= \frac{6.62 \times 10^{-34}}{3.6 \times 10^{-10} \times 3 \times 10^8}$$

$$= 6.135 \times 10^{-33} \text{ kg}$$

39. Ans (3)

$$\text{Number of atoms} = n \times N_A \times \text{atomicity}$$

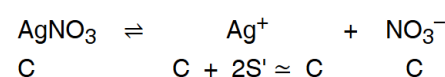
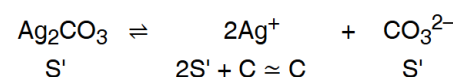
41. Ans (3)

$$\Delta G^\circ = -2.303 RT \log K_p$$

$$= -2.303 \times \frac{8.314}{1000} \times 298 \log 2 \times 10^{-29}$$

$$= +163.7 \text{ kJ}$$

43. Ans (2)



$$K_{sp} = (\text{C})^2 \times \text{S}'$$

$$\text{S}' = \frac{K_{sp}}{\text{C}^2} = \frac{8 \times 10^{-12}}{(0.1)^2} = 8 \times 10^{-10}$$

46. Ans (2)

NCERT XI Pg. # 7

47. Ans (3)

NCERT XI Pg. # 7

48. Ans (3)

NCERT-XI Pg # 18

49. Ans (4)

NCERT Pg # 13

50. Ans (3)

NCERT-XI, Pg.# 30-33

51. Ans (3)

NCERT XI - Pg. # 32-33

52. **Ans (3)**
NCERT-XI Pg. # 27
Ectocarpus, Dictyota, Sargassum, Fucus
53. **Ans (3)**
NCERT XI - Pg. # 31
54. **Ans (4)**
b, c and d are correct
NCERT-XI, Pg # 33
55. **Ans (1)**
NCERT-XI, Pg. # 87
56. **Ans (2)**
XI NCERT, Pg. # 100
57. **Ans (3)**
NCERT XI, Pg. # 121
58. **Ans (4)**
NCERT-XI, Pg # 124
59. **Ans (1)**
NCERT-XI, Pg # 123
60. **Ans (4)**
NCERT-XI, Pg # 124
61. **Ans (4)**
NCERT-XI, Pg. # 125,126,127
63. **Ans (1)**
NCERT-XI, Pg. # 105, 110
64. **Ans (4)**
NCERT-XI, Pg. # 110, 111
65. **Ans (2)**
NCERT-XI, Pg. # 118
66. **Ans (2)**
NCERT-XI, Pg # 117
67. **Ans (1)**
NCERT-XI, Pg. # 147
68. **Ans (1)**
NCERT-XI, Pg. # 138, 147, 149
69. **Ans (3)**
NCERT-XI, Pg. # 149

70. **Ans (3)** Telegram: @NEETxNOAH
NCERT-XI, Pg. # 146
71. **Ans (4)**
NCERT-XI, Pg. # 140
72. **Ans (4)**
NCERT-XI, Pg. # 156, 158, 160
73. **Ans (2)**
NCERT XI, Pg. # 158, 160
74. **Ans (4)**
NCERT XI, Pg. # 162
75. **Ans (1)**
NCERT XI Pg # 160
76. **Ans (1)**
NCERT XI Pg # 175-178
77. **Ans (4)**
NCERT-XI, Pg. # 174
78. **Ans (2)**
NCERT-XI, Pg. # 176
79. **Ans (1)**
XI NCERT Pg. # 14, 15
80. **Ans (2)**
NCERT, Pg. # 4
81. **Ans (3)**
NCERT-XI Pg. # 26
82. **Ans (4)**
NCERT-XI, Pg. # 90
84. **Ans (2)**
NCERT (XI) Pg # 91
85. **Ans (3)**
NCERT XI Pg. # 99
86. **Ans (1)**
NCERT (XI) Pg # 126
87. **Ans (3)**
NCERT XI, Pg. # 107

88. Ans (4)

NCERT-XI, Pg. # 142

89. Ans (3)

NCERT XI, Pg. # 163

90. Ans (3)

NCERT-XI, Pg. # 159, 160

91. Ans (1)

NCERT XI Pg # 54

92. Ans (4)

NCERT XI Pg # 54

99. Ans (1)

NCERT XIth Pg. No. # 114, 5th Para

102. Ans (1)

NCERT-XI Pg. # 116

120. Ans (2)

XI NCERT Pg. No :- 321

122. Ans (2)

NCERT-XII, Pg. # 247-248

123. Ans (2)

NCERT-XII, Pg. # 247-248

124. Ans (3)

NCERT Pg. # 245

125. Ans (4)

NCERT Pg. # 245

126. Ans (4)

NCERT Pg. # 243-244

127. Ans (4)

NCERT Pg. # 246

128. Ans (4)

NCERT Pg. # 243

129. Ans (1)

NCERT (old) Page No.: 312

136. Ans (4)

$$\text{Gravitational constant, } G = 6.67 \times 10^{-7} \frac{\text{N} \cdot \text{m}^2}{\text{kg}^2}$$

$$[G] = [M^{-1} L^3 T^{-2}]$$

Gravitational constant has dimensions and constant value.

137. Ans (3)

$\frac{B^2}{\mu_0}$ indicate energy density and $\epsilon_0 E^2$ also indicate energy density.

$$\Rightarrow \frac{B^2}{\mu_0 \epsilon_0 E^2} \text{ is dimensionless } [M^0 L^0 T^0 A^0]$$

138. Ans (1)

5000 has only one SF i.e. 5

statement-2 is also true as a number can have any number of significant digits.

139. Ans (1)

$$R = \frac{V}{I} = \frac{10}{5} = 2\Omega$$

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I} = \frac{0.1}{10} + \frac{0.1}{5}$$

$$\Delta R = \frac{0.3}{10} \times 2 = 0.06 \Omega$$

$$\Rightarrow R = (2 \pm 0.06) \Omega$$

140. Ans (1)

$$1 \text{ MSD} = x \text{ cm}$$

$$LC = 1 \text{ MSD} - 1 \text{ VSD}$$

$$y \text{ VSD} = (y - 1) \text{ MSD} \Rightarrow 1 \text{ VSD} = \left(1 - \frac{1}{y}\right) \text{ MSD}$$

$$\Rightarrow LC = 1 \text{ MSD} \left[1 - \left(1 - \frac{1}{y}\right)\right] = \frac{1 \text{ MSD}}{y}$$

$$LC = \frac{x}{y} \text{ cm}$$

141. Ans (3)

$$\text{Reading} = (6) (1 \text{ mm}) + 5(0.1 \text{ mm})$$

$$= 6.5 \text{ mm}$$

$$\text{True value} = 6.5 \text{ mm} - \text{zero error}$$

$$= 6.5 - 0.2$$

$$= 6.3 \text{ mm}$$

142. Ans (1)

$$\text{Least count} = \frac{1}{100} = 0.01 \text{ mm}$$

$$\text{Zero error} = + (15) (0.01) = 0.15 \text{ mm}$$

$$\begin{aligned} \text{Correct Reading} &= (2.00 + 35 \times 0.01) - 0.15 \text{ mm} \\ &= 2.20 \text{ mm} \end{aligned}$$

143. Ans (2)

$$w = mgS$$

$$w \propto S \text{ (mg = const.)}$$

$$S_2^{\text{nd}} : S_3^{\text{rd}} : S_4^{\text{th}} = 3 : 5 : 7$$

$$\therefore w_2^{\text{nd}} : w_3^{\text{rd}} : w_4^{\text{th}} = 3 : 5 : 7$$

144. Ans (4)

NCERT Pg # 47

$$-\int_{10}^{20} \frac{2}{y} dy = KE_F - \frac{1}{2} \times 1.2 \times (4)^2$$

$$+ 2[\ell ny]_{10}^{20} = -KE_F + 9.6$$

$$KE_F = 9.6 - 2\ln 2 \Rightarrow 9.6 - 1.4$$

$$KE_F \Rightarrow 8.2 \text{ J}$$

145. Ans (2)

For collision

$$(u_y)_A = (u_y)_B$$

$$\frac{u}{\sqrt{2}} = 50 \sin 53^\circ$$

$$u = 40\sqrt{2} \text{ m/s}$$

146. Ans (4)

$$\vec{v}_i = -10\hat{j}$$

$$\vec{v}_f = -10\hat{i}$$

$$\vec{\Delta v} = \vec{v}_f - \vec{v}_i$$

$$= -10\hat{i} + 10\hat{j}$$

147. Ans (1)

$$\tan \theta = \frac{4H}{R}$$

$$\tan 45^\circ = \frac{4H}{400}$$

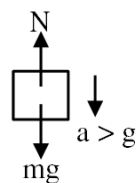
$$H = 100 \text{ m}$$

148. Ans (2)

$$H_{\text{max}} = \frac{R_{\text{max}}}{2} = 40 \text{ m}$$

149. Ans (4)

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$$N = 0$$

150. Ans (2)

$$N_1 = 30, \quad f_1 = 12$$

$$N_2 = 50, \quad f_2 = 20$$

$$\rightarrow a = \frac{40-32}{8} = 1 \text{ m/s}^2$$

$$\rightarrow a = 1 \text{ m/s}^2$$

$$T - 12 = 3(1)$$

$$T = 15 \text{ N}$$

151. Ans (3)

NCERT, Pg. # 99

The velocity of COM remains constant if no external force act on the system.

152. Ans (4)

(NCERT Pg. No. 95)

(a) let $m_R = m$ then $m_P = m_Q = 3m$

$$X_{\text{cm}} = \frac{(3m)(0) + (3m)(0) + m(2)}{3m + 3m + m}$$

$$X_{\text{cm}} = 2/7$$

(b) let $m_P = m_Q = m_R = m$

$$X_{\text{cm}} = \frac{m(0) + m(0) + m(2)}{m + m + m} = \frac{2}{3}$$

(c) let $m_P = m_Q = m$

$$m_R = \frac{m + m}{4} = \frac{m}{2}$$

$$X_{\text{cm}} = \frac{m(0) + m(0) + \frac{m}{2}(2)}{m + m + \frac{m}{2}} = \frac{2}{5}$$

153. Ans (4)

(NCERT Pg. no. 100)

When a bomb at rest explodes, the COM of the fragments remains at rest. because forces acting on fragments during explosion are internal forces and internal forces can not change the state of motion of COM.

154. Ans (1)

NCERT Pg # 42

$$v = \omega R$$

$$v = \omega \times 0 = 0$$

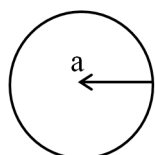
155. Ans (4)

In uniform circular motion -

$v = \text{speed} = \text{constant}$

$$K.E. = \frac{1}{2} mv^2 = \text{constant}$$

acceleration towards centre.



156. Ans (4)

$$T_{\max} = T_A = \frac{mV_A^2}{\ell} + mg$$

$$T_{\min} = T_B = \frac{mV_B^2}{\ell} - mg$$

157. Ans (4)

NCERT Pg. # 105

$$\tau = I\alpha$$

$$\boxed{\alpha = 0}$$

158. Ans (4)

$$I = \frac{2}{5} Mr^2 = \frac{2}{5} \times \left(\frac{4}{3} \pi r^3 \rho \right) r^2$$

$$\Rightarrow I \propto r^5 \text{ as } \rho = \text{constant.}$$

159. Ans (3)

$$\frac{GM}{R^3} \left(\frac{R}{4} \right) = \frac{GM}{(R+d)^2} \Rightarrow d = R$$

160. Ans (2)

Moving along tangential direction.

161. Ans (1)

$$TE = KE + PE$$

$$E_0 + \left(\frac{-E_0}{2} \right) = \frac{E_0}{2} = +ve$$

So as $TE = +ve$ so system is unbounded and hence particle will escape from earth.

Also by energy conservation, energy at $\infty = \text{positive}$

$$\Rightarrow KE \text{ at } \infty = +ve$$

$$\Rightarrow \text{speed at } \infty = +ve$$

162. Ans (4)

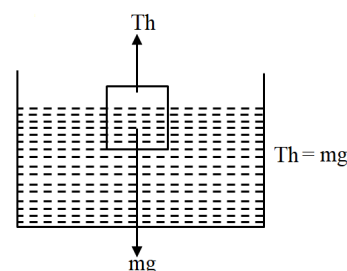
Excess pressure inside liquid drop

$$P_{\text{ex}} = \frac{2T}{R} = \frac{4T}{D} \quad \because D_1 = 3 \text{ cm}, D_2 = 2.5 \text{ cm}$$

$$\text{So, } P_{\text{ex}_1} = \frac{4T}{3\text{cm}} \text{ \& } P_{\text{ex}_2} = \frac{4T}{2.5 \text{ cm}}$$

$$\Rightarrow \frac{P_{\text{ex}_1}}{P_{\text{ex}_2}} = \frac{2.5}{3} = \frac{25}{30} = \frac{5}{6}$$

163. Ans (1)



164. Ans (2)

$$\rho_1 h_1 = \rho_2 h_2$$

$$50 \times 200 = 25 \times \rho_2$$

$$400 \text{ kg/m}^3 = \rho_2$$

165. Ans (3)

$$n_{\text{mix}} = n_1 + n_2$$

$$\frac{P_{\text{mix}} V_{\text{mix}}}{RT} = \frac{P_1 V_1}{RT} + \frac{P_2 V_2}{RT}$$

$$P_{\text{mix}} (1) = 1(1) + (0.5) (2)$$

$$P_{\text{mix}} = 2 \text{ atm}$$

166. Ans (2)

$$\text{Energy} = \frac{nfRT}{2} = \frac{(1)fRT}{2} = \frac{fRT}{2}$$

$$\text{Atomicity} = 1 \quad \text{Energy} = \frac{3RT}{2}$$

$$\& f = 3$$

$$\text{Atomicity} = 2 \quad \text{Energy} = \frac{5RT}{2}$$

$$\& f = 5$$

$$\text{Atomicity} = 3 \quad \text{Energy} = \frac{6RT}{2} = 3RT$$

$$\& f = 6$$

167. Ans (2)

$$V_{\text{rms}} = \sqrt{\frac{3RT}{M_W}}, V_{\text{avg}} = \sqrt{\frac{8RT}{\pi M_W}}$$

$$\therefore \frac{V_{\text{avg}}}{V_{\text{rms}}} = \sqrt{\frac{8}{3\pi}}$$

$$\Rightarrow V_{\text{avg}} = 1260 \times 0.921$$

$$\approx 1161 \text{ m/s}$$

168. Ans (3)

(1) $PV = C$, $P \propto \frac{1}{V}$ Correct

(2) $PV = C$,

$$\log P + \log V = \log C$$

$$\log P = -\log V + \log C$$

–Ve slope and + Ve intercept correct

(3) Wrong

(4) $PV = C$ correct

169. Ans (1)

$$\text{Slope} \propto \frac{1}{P} \quad PV = nRT$$

$$\tan 30^\circ \propto \frac{1}{P_1} \dots \dots \dots (1)$$

$$\tan 60^\circ \propto \frac{1}{P_2} \dots \dots \dots (2)$$

$$(1) \div (2) \quad \frac{P_2}{P_1} = \frac{\tan 30^\circ}{\tan 60^\circ} = \frac{1}{\sqrt{3} \times \sqrt{3}}$$

$$P_1 = 3P_2$$

170. Ans (3)

$$f_{\text{mix}} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2}$$

$$5 = \frac{N(6) + 2(3)}{N + 2}$$

$$\Rightarrow 5N + 10 = 6N + 6 \Rightarrow \boxed{N = 4}$$

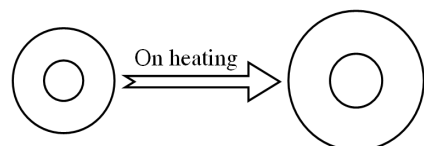
171. Ans (2)

$$(\Delta L)_A = (\Delta L)_B$$

$$\alpha_A L_A \Delta T = \alpha_B L_B \Delta T$$

$$\frac{L_A}{L_B} = \frac{\alpha_B}{\alpha_A}$$

172. Ans (1)



173. Ans (4)

Telegram: @NEETxNOAH

$$\text{For mixture :- } C_V = \frac{n_1 C_{V1} + n_2 C_{V2}}{n_1 + n_2}$$

$$\text{and } C_P = \frac{n_1 C_{P1} + n_2 C_{P2}}{n_1 + n_2}$$

$$\text{For He : } n_1 = 4, f = 3, C_{V1} = \frac{3R}{2}, C_{P1} = \frac{5R}{2}$$

$$\text{For O}_2 \text{ :- } n_2 = 1/2, f = 5, C_{V2} = \frac{5R}{2}, C_{P2} = \frac{7R}{2}$$

$$\therefore \frac{C_P}{C_V} = 1.62$$

174. Ans (1)

Assertion is definition of ideal black body hence correct. Reason is also correct as per Wein's Law.

175. Ans (3)

From the conservation of mechanical energy

$$\frac{1}{2} kx^2 = \frac{1}{2} mv^2$$

$$x = v \sqrt{\frac{m}{k}}$$

176. Ans (3)

$$V = a$$

$$\omega \sqrt{a^2 - x^2} = \omega^2 x$$

$$a^2 - x^2 = \omega^2 x^2$$

$$(3)^2 - (2)^2 = \omega^2 (2)^2$$

$$\omega^2 = \frac{5}{4} \quad \omega = \sqrt{\frac{5}{4}}$$

$$T = 2\pi \sqrt{\frac{4}{5}} = \frac{4\pi}{\sqrt{5}}$$

177. Ans (3)

$$k_{\text{eq}} = k + 2k = 3k$$

$$T = 2\pi \sqrt{\frac{m}{k_{\text{eq}}}} = 2\pi \sqrt{\frac{m}{3k}}$$

$$n = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{3k}{m}}$$

178. Ans (4)

$$2d = v \times t$$

$$d = \frac{v \times t}{2} = \frac{340 \times 1}{2} = 170 \text{ m}$$

180. Ans (4)

$$n = \frac{1}{2\ell} \sqrt{\frac{T}{\pi r^2 \rho}} = \frac{1}{2\ell r} \sqrt{\frac{T}{\pi \rho}}$$

$$\frac{n_1}{n_2} = \frac{\ell_2 r_2}{\ell_1 r_1}, \quad \frac{n}{n_2} = \frac{2}{1} \times \frac{2}{1}$$

$$n_2 = \frac{n}{4}$$